
EEE6778. Applied Machine Learning II



Module 2. Convolutional Neural Networks

Computer Vision
Convolutional Filters
Example Network Architectures

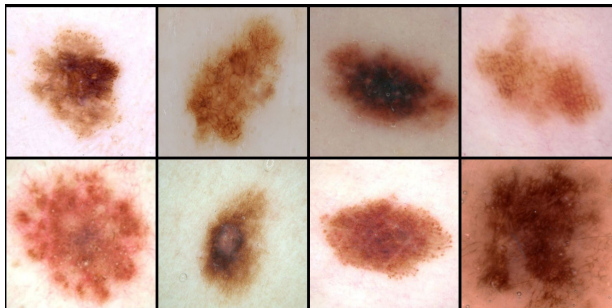


Computer Vision

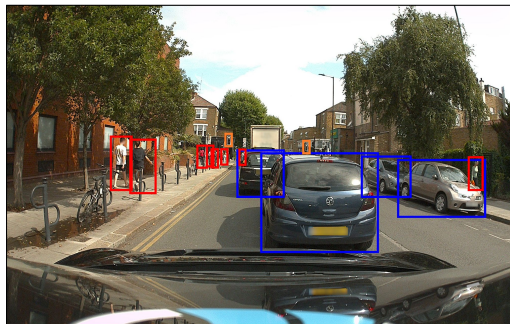
ML Applications of Computer Vision

“Computer vision was one of the first fields to be transformed by the deep learning revolution, predominantly thanks to the CNN architecture.” (Bishop, 2024)

Classification



Detection



Segmentation



+ Caption generation from image, synthesis of new images, super-resolution, etc!

Image Data

An image consists of a rectangular array of pixels, each with:

- A gray-scale intensity or
- A triplet of red, green, and blue (RGB) channels.

Pixel intensities are non-negative and depend on the camera or sensor.

Typically represented as finite precision values (e.g., 8-bit integers ranging from 0 to 255).

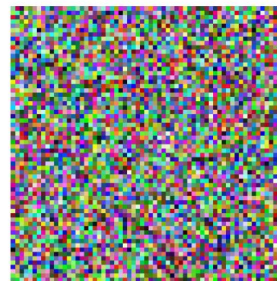
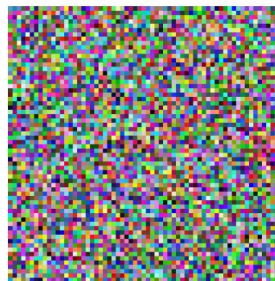
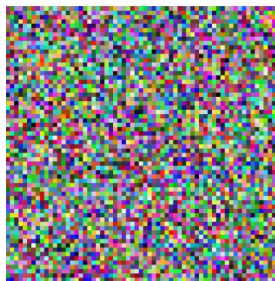
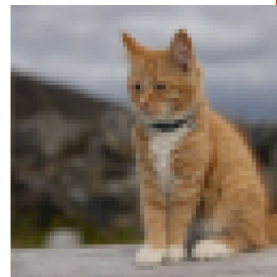
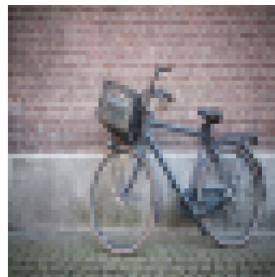
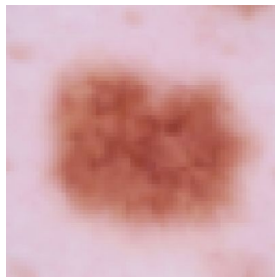
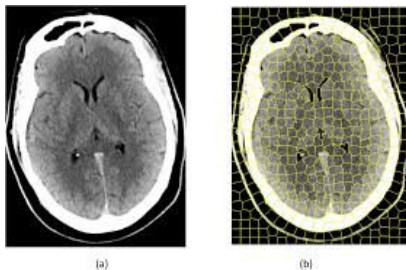


Image Data

Types of Image Data:

- **Medical Imaging:** MRI scans use three-dimensional grids of voxels.
- **Video Sequences:** Treated as three-dimensional structures where frames are stacked over time.



Challenges in Applying Neural Networks:

- **High Dimensionality:** Cameras capture megapixel images, leading to a vast amount of data.
- **Unstructured Representation Issues:**
 - Ignoring image structure can lead to infeasible models.
 - Relative positions of pixels contain essential information.

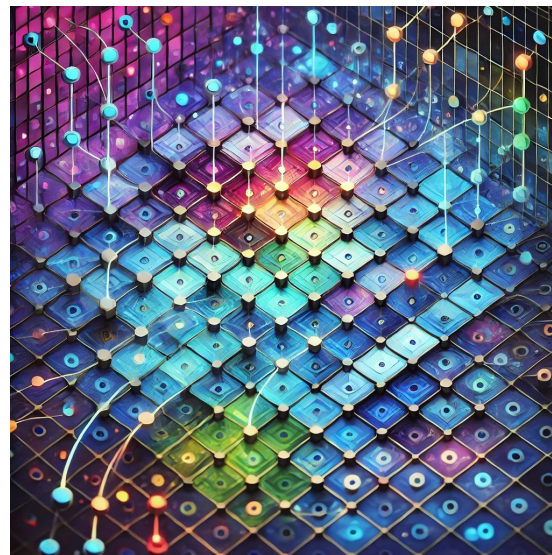
Image Data - Importance of Spatial Relationships

Local Correlations Matter:

- In natural images, nearby pixels are more likely to have similar colors.
- This knowledge can be leveraged to improve neural network performance.

Encoding Spatial Knowledge in Neural Networks:

- Helps reduce the number of parameters.
- Leads to improved generalization and model accuracy.

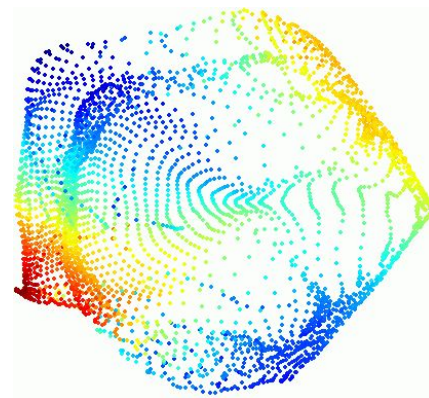


AI-generated image using DALL-E

Challenges in Traditional Neural Networks for Image Data

High Dimensionality Issue

- Images have a large number of pixels (e.g., a $10^3 \times 10^3$ color image has $\sim 10^6$ pixels).
- Fully connected networks would require an extremely high number of weights (e.g., 10^9 hidden units $\rightarrow 3 \times 10^9$ weights in the first layer).
- This leads to massive data storage requirements and computational inefficiencies.



Challenges in Traditional Neural Networks for Image Data

Need for an Efficient Architecture

- A naive network would need to learn invariances and equivalences (e.g., recognizing a rotated or shifted object as the same).
- Directly learning all variations requires large datasets, making training inefficient.
- Better approach: Use an architecture that incorporates inductive biases to leverage known structures in images, improving generalization and reducing data requirements.

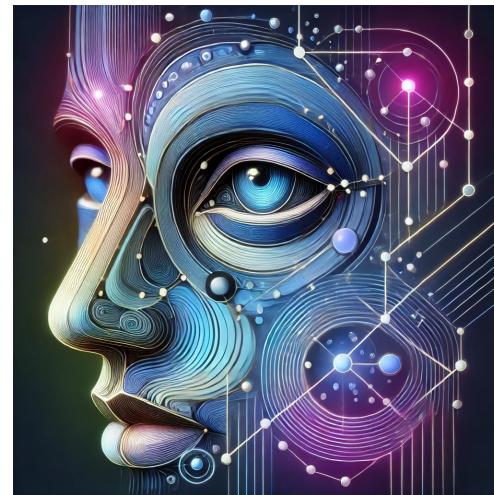
Key Inductive Biases for Image Data in Neural Networks

Hierarchy

- Images naturally contain hierarchical structures (e.g., a face consists of eyes, a nose, and a mouth, which in turn contain finer details like pupils and nostrils).
- Learning should be layered, capturing lower-level features first and composing them into higher-level features

Locality

- Features in images are often local (e.g., edges, textures, corners).
- A model should recognize a feature within a small region rather than needing a fully connected approach.

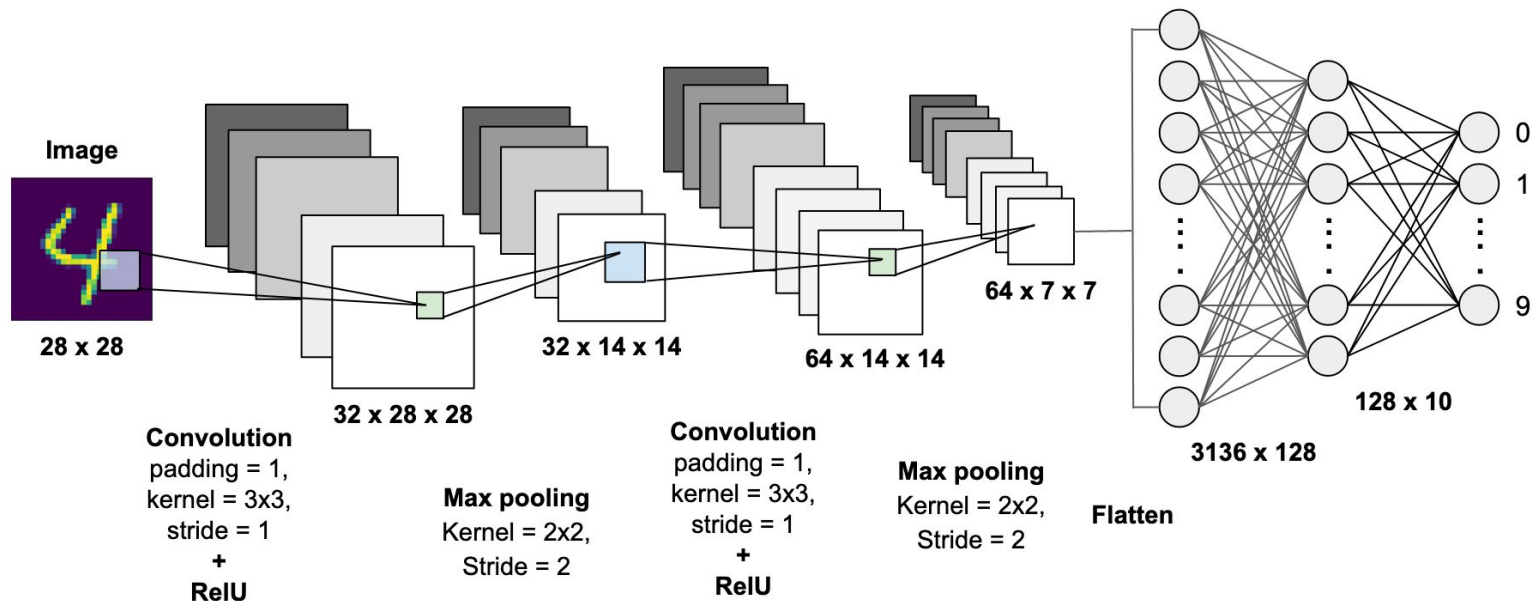


Convolutional Neural Networks (CNNs) naturally exploit these properties through shared filters and pooling operations.



Convolutional Filters

CNN - Big Picture



<https://towardsdatascience.com/mnist-handwritten-digits-classification-using-a-convolutional-neural-network-cnn-af5fafbc35e9>

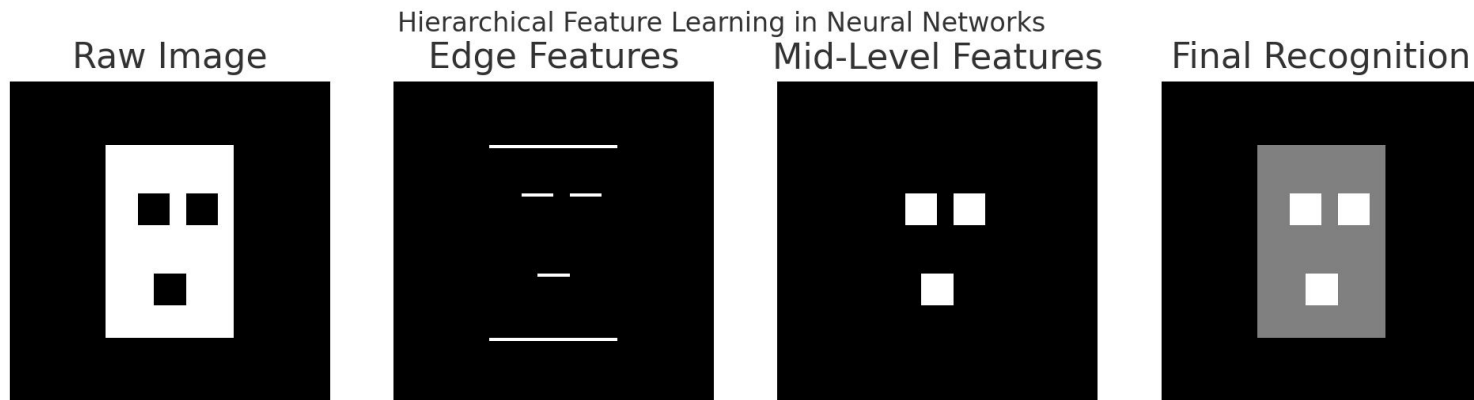
Hierarchical Feature Learning in Deep Learning

Learning Features at Multiple Levels

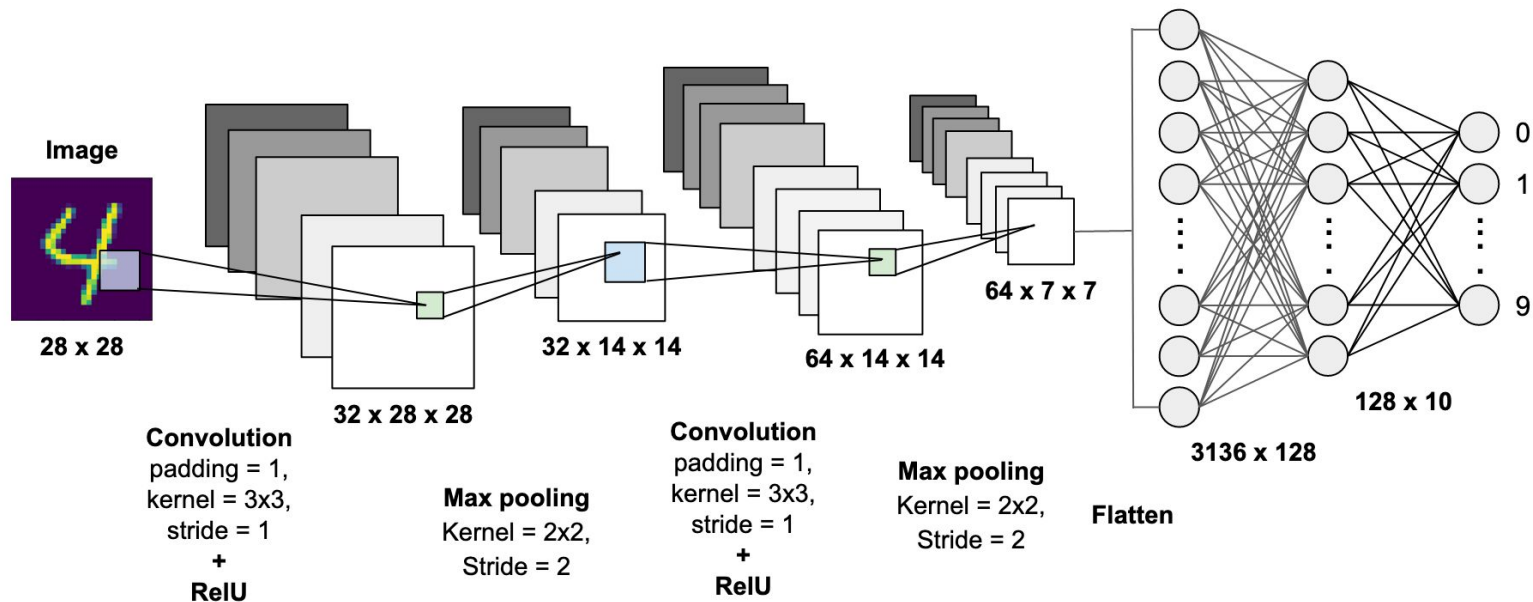
Lower layers learn edges, textures, and shapes.

Higher layers learn object parts and entire objects.

Complex structures are composed from simpler learned features.



CNN - Big Picture



<https://towardsdatascience.com/mnist-handwritten-digits-classification-using-a-convolutional-neural-network-cnn-af5fafbc35e9>

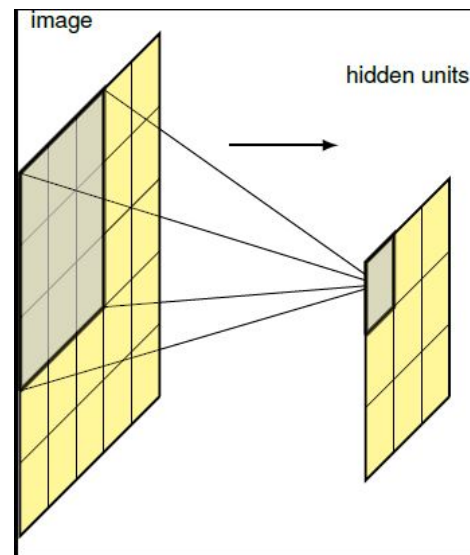
Convolutional Filters - Receptive Fields

- Are a small rectangular patch of pixel values from the image.
- Defines the local area that a neural network unit processes.
- Captures the notion of locality, ensuring nearby pixels contribute to feature detection.
- The output is a weighted linear combination of the receptive field's pixel values.

A nonlinear activation function (ReLU) transforms the result:

$$z = \text{ReLU}(\mathbf{w}^T \mathbf{x} + w_0)$$

- $\mathbf{x}$: Vector of pixel values in the receptive field.
- $\mathbf{w}$: Weights associated with each input.
- w_0 : Bias term.
- ReLU activation function ensures non-linearity in learning.



0.4	1.7	0.9
2.3	-2.1	4.0
-1.4	0.7	2.1

(Bishop, 2024)

Convolutional Filters - Receptive Fields

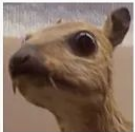
The most successful non-linearity for CNN's is the Rectified Non-Linear unit (ReLU), which combats the vanishing gradient problem occurring in sigmoids. ReLU is easier to compute and generates sparsity (not always beneficial).

Convolutional Filters or Kernels

- A filter (kernel) is a small two-dimensional grid of weights applied to an image.
- Can be visualized as an image itself.
- Helps detect local patterns such as edges, textures, or specific features.

Examples of kernel filters for CNN's.

Edge detection



$$* \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix} =$$

Kernel



Sharpen

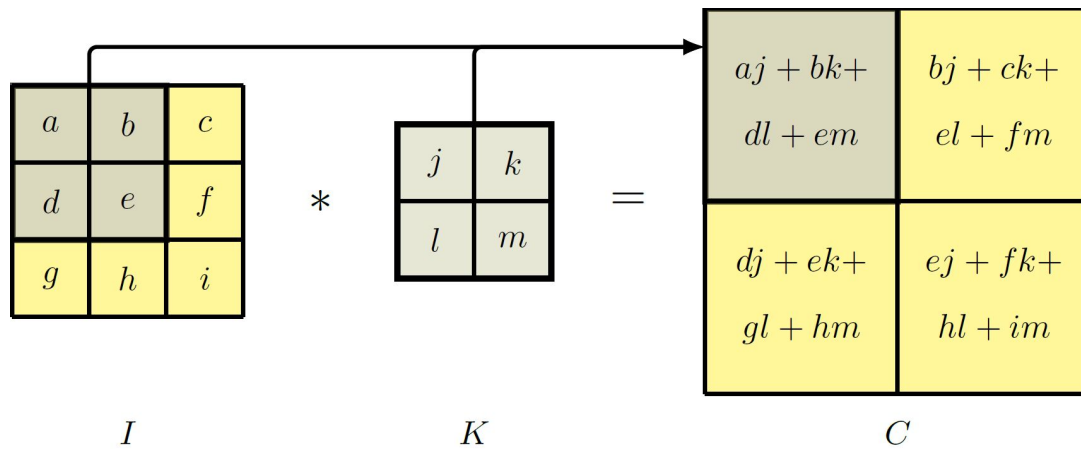


$$* \begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix} =$$

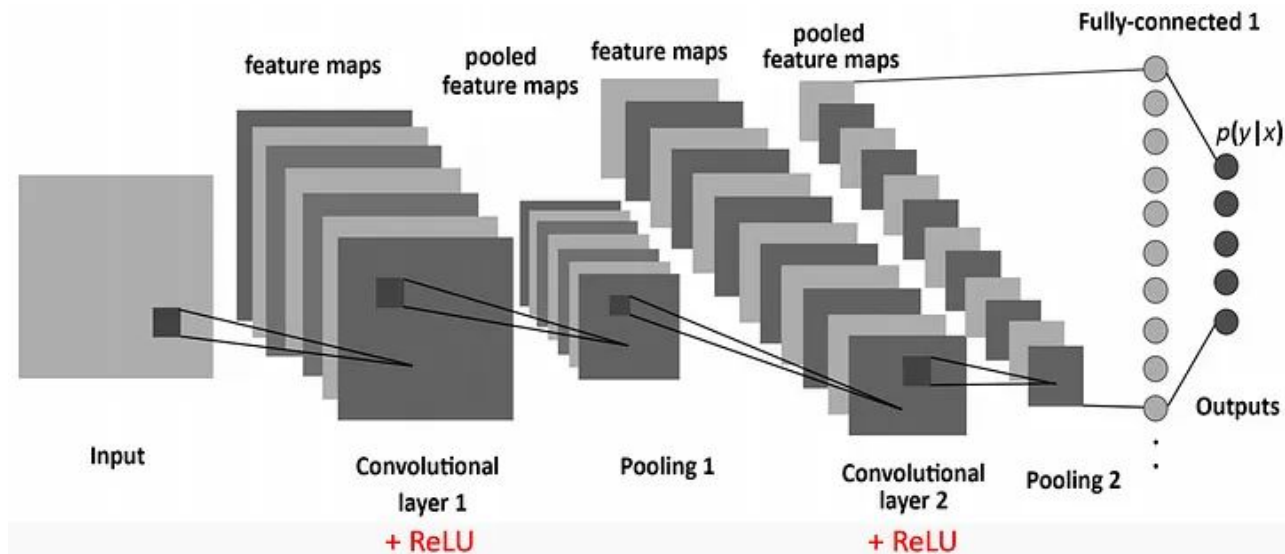

Convolutional Filters - Feature Maps

- After filters pass over an image, each filter produces a feature map.
- These feature maps capture detected patterns, such as edges, textures, or objects.

Example of a 3 x 3 image convolved with a 2 x 2 filter to give a resulting 2 x 2 feature map



Convolutional Filters - Feature Maps

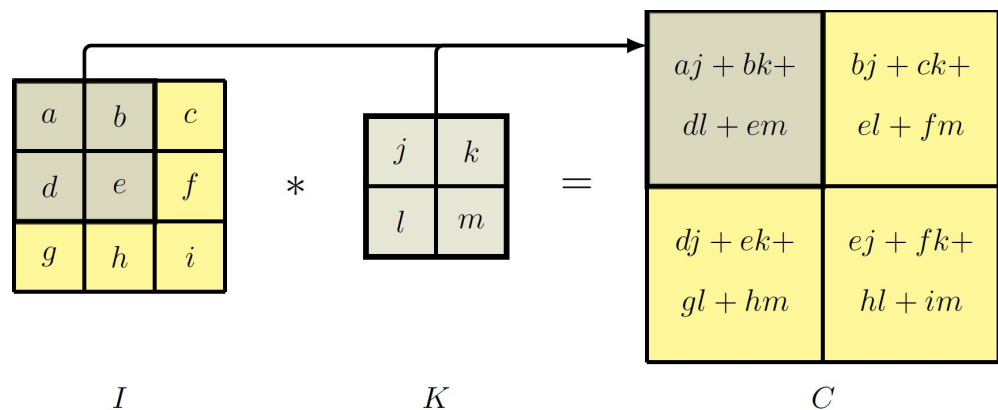


<https://towardsdatascience.com/simple-introduction-to-convolutional-neural-networks-cdf8d3077bac>

Convolutional Filters - Feature Maps

Mathematical Definition of Convolution

- For an input image $I(j, k)$ and a filter (kernel) $K(l, m)$, feature map is computed as $C(j, k) = \sum_l \sum_m I(j + l, k + m)K(l, m)$
- This operation slides the filter over the image and computes weighted sums at each position.





Convolutional Filters - Feature Maps

Concept of Equivariance

- If a feature appears in a specific location in an image, it will also appear in the same relative location in the feature map.
- This spatial consistency is an example of equivariance in convolutional neural networks.
- The network maintains this property due to the shared weights across hidden units.

Sparse Connections in Convolutional Layers

- Unlike fully connected layers, convolutions create sparse connections where only local pixel neighborhoods influence outputs.
- This reduces computational complexity and improves feature detection efficiency.

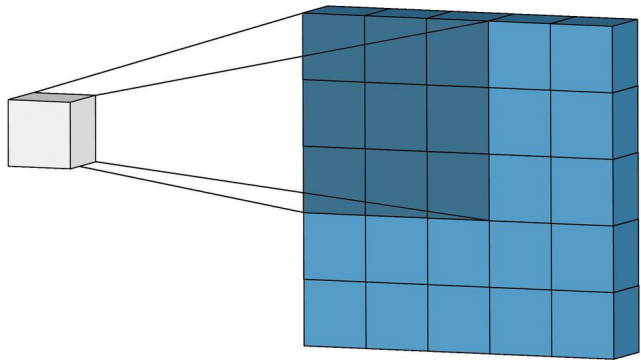
Batch Normalization in Convolutional Networks

- When using batch normalization, the same mean and variance are applied across spatial locations in the feature map.
- This ensures consistency while normalizing activations in CNNs.

Convolutional Filters - Recap

We know that MLPs:

- Do not scale well for images
- Ignore the information brought by pixel position and correlation with neighbors
- Cannot handle translations



The general idea of CNN's is to intelligently adapt to the properties of images:

- Pixel position and neighborhood have semantic meanings
- Elements of interest can appear anywhere in the image

Moving Forward

Explore the CNN Explainer before next class!

